1. Introduction

Purpose of Database Systems

- **Definition (exam style):**
  A database system is designed to efficiently store, manage, and retrieve large amounts of data while ensuring data consistency, security, and ease of access.
- **Simple words:** Before databases, people used file systems which caused problems like redundancy (same data stored multiple times), inconsistency, difficulty in accessing, and poor security. Databases solve these issues.

View of Data

- **Definition:**
  The way data is perceived by different users at different levels of abstraction: physical level, logical level, and view level.
- **Simple words:** Think of it like layers – the engineer sees raw storage, the designer sees tables and relationships, the end user just sees an app screen.

Data Models

- **Definition:**
  Data models are frameworks that describe how data is structured, stored, and manipulated. Examples: Relational model, Entity-Relationship model, Object-oriented model.
- **Simple words:** A data model is like a blueprint of how data is arranged in a database.

Database Language

- **Definition:**
  DBMS provides specialized languages to define, manipulate, and control data.
    - **DDL (Data Definition Language):** Defines structure (CREATE, ALTER).
    - **DML (Data Manipulation Language):** Manipulates data (INSERT, UPDATE, DELETE).
    - **DCL (Data Control Language):** Manages permissions (GRANT, REVOKE).

    - **TCL (Transaction Control Language):** Handles transactions (COMMIT, ROLLBACK).
- **Simple words:** SQL is the main language of databases, divided into parts for different tasks.

Transaction Management

- **Definition:**
  Transaction management ensures that database operations are executed completely or not at all, maintaining consistency, using the **ACID properties** (Atomicity, Consistency, Isolation, Durability).
- **Simple words:** A transaction is like an ATM withdrawal – it must either happen fully or not at all.

Storage Management

- **Definition:**
  It is the process of storing data efficiently on physical storage devices while providing fast access and data protection.
- **Simple words:** It's how the DBMS organizes data on hard drives or SSDs.

Database Administrator (DBA)

- **Definition:**
  A DBA is a person responsible for managing, controlling, and maintaining the database system, including security, backup, and performance.
- **Simple words:** The "caretaker" of the database.

Database Users

- **Definition:**
  People who interact with the database system. Types:
    - **End users** (casual, naive, sophisticated, standalone).
    - **Application programmers.**

    - **System analysts.**
- **Simple words:** Different people use the DB differently, from students searching results to admins managing data.

Overall System Structure

- **Definition:**
  The DBMS architecture includes **users → DBMS software → database storage**. It manages interaction between applications and stored data.
- **Simple words:** Think of it as a layered structure where users send requests → DBMS processes them → database stores answers.

Classification of Database Management Systems

- **Definition:**
  DBMS can be classified based on:
    - **Data model** (Relational, Hierarchical, Network, Object-oriented).
    - **Number of users** (single-user, multi-user).
    - **Distribution** (centralized, distributed, client-server).
- **Simple words:** Just like apps are categorized, DBMS are grouped by how they handle data.

Three-Schema Architecture

- **Definition:**
  A DBMS architecture that separates database structure into **three levels**:
    - **Internal level:** physical storage details.
    - **Conceptual level:** logical structure of data.
    - **External level:** different user views.
- **Simple words:** It's like having different "views" of the same data for different people (engineer, manager, customer).

---

2. Entity–Relationship (E-R) Model

Basic Concepts

- **Definition:**
  The E-R model represents real-world entities, their attributes, and relationships between them.
- **Simple words:** It's like a diagram showing things (entities) and how they are related.

Design Issues

- **Definition:**
  Problems to consider while designing E-R diagrams, such as: choosing entity vs. attribute, relationship degree, key selection, redundancy removal.
- **Simple words:** Things to watch out for so your diagram is correct and efficient.

Mapping Constraints

- **Definition:**
  Constraints that describe how entities are related:
  - **1:1 (one to one)**
  - **1:N (one to many)**
  - **M:N (many to many)**
- **Simple words:** Rules about how many entities can be connected to each other.

Keys

- **Definition:**
  Keys uniquely identify entities.
  - **Primary key, Candidate key, Foreign key, Super key.**
- **Simple words:** Keys are like ID cards for database records.

E-R Diagram

- **Definition:**
  A visual representation of entities, attributes, and relationships.
- **Simple words:** A flowchart for data.

Weak Entity Sets

- **Definition:**
  An entity that cannot be uniquely identified without a "strong" entity and uses a **partial key**.
- **Simple words:** A dependent entity (e.g., "Dependent" of "Employee").

Extended E-R Features

- **Definition:**
  Advanced features like:
  - **Generalization** (common to specific).
  - **Specialization** (specific to common).
  - **Aggregation** (relationship as entity).
- **Simple words:** Extra tools for more complex diagrams.

Design of an E-R Database Schema

- **Definition:**
  The process of converting requirements into an E-R diagram and then into relational schema.
- **Simple words:** Turning real-world problems → diagram → database tables.

Reduction of E-R Schema to Tables

- **Definition:**
  The method of transforming an E-R model into relational tables using mapping rules.

- **Simple words:** Converting diagrams into actual tables in the database.

---

3. Relational Model

Structure of Relational Databases

- **Definition:**
  Data is organized into relations (tables), with rows (tuples) and columns (attributes).
- **Simple words:** The most common type of database, like Excel sheets with rules.

Relational Algebra

- **Definition:**
  A procedural query language that uses operations (SELECT, PROJECT, JOIN, UNION, INTERSECT, etc.) to manipulate relations.
- **Simple words:** A set of mathematical operations to work with tables.

Tuple Relational Calculus (TRC)

- **Definition:**
  A non-procedural query language where queries are expressed using variables that represent tuples.
- **Simple words:** You specify **what you want**, not how to get it (based on tuples).

Domain Relational Calculus (DRC)

- **Definition:**
  Similar to TRC, but queries are expressed using domain variables (attribute values instead of whole tuples).
- **Simple words:** Instead of talking about full rows, you talk about individual columns.